

interfaces, Ethernet being the *de facto* standard, are ubiquitous. For long, the world relied on the 100Mbps standard. Then, gigabit options started arriving.

"The most pervasive networking technology has evolved from operating at moderate speeds of 10 Mbps (10 BaseT) and 100 Mbps (100 BaseTX), and now to Gigabit Ethernet (GbE) of four lanes at 250 Mbps (1000 BaseT) and 10 Gbps. At 10 Gbps, four XAUI electrical lanes operate at 3.125 Gbps," remarks Evan Sun, technical marketing manager, Tektronix-Asia Pacific Region.

Today, there are 100Gbps network interfaces available commercially, the first having been introduced by Juniper networks a little over a year ago. While such high-end Ethernet interfaces are being used only in enterprise servers and industrial equipment now, it is not rare to see discussions and debates about when our home network will become this fast!

Ugly wires, disappear please

Speak of any form of communication, and wireless technology is sure to feature! We have already mentioned Wireless USB as one of the players in this space. However, when it comes to wireless local-area networking, the 802.11 family of standards (commercially known as Wi-Fi) rules the space. Today, you have devices ranging from printers to televisions that are Wi-Fi-capable.

The most recent addition to this clique is the 802.11n standard published in October 2009. It adds multiple-input-multiple-output (MIMO) antennae and other such interesting features to build upon the earlier 802.11 standards. It increases almost all the critical aspects of a wireless network including speed, range and compatibility. A year after its ratification, we can safely say that the standard does live up to the claims to a large extent.

Last year, another popular wireless protocol went for an upgrade. The new Bluetooth v4.0 comprises three sub-versions that feature high speed, low

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energy consumption and longer range, respectively. These specifications can be used separately or in-sync with each other in the same device. Version 4.0 borrows from Wi-Fi technology to increase the speed and range to 24 Mbps and 61 metres (200 feet).

There are various other types of wireless interfaces, such as those used by smartcard readers and cards to communicate with each other. There is a lot of research happening to increase the data transfer speeds of these interfaces, since contact-less smartcards are now being used for biometric data transfer, multimedia file downloads, etc. Here is an example of one such research venture. CEA's Laboratory for Electronics and Information Technology (CEA-Leti), a French research organisation, has created a prototype of a complete phase modulation system, a reader and a card that now achieves speeds of 6.8 Mbps. Phase modulation's spectral characteristics are superior to those of amplitude modulation, allowing considerably higher speeds than the current limit of 848 kbps. Leti's current work on optimisation of signal processing aims to attain speeds greater than 10 Mbps.

Multimedia interfaces join the race too!

With the advent of HD televisions, set-top boxes and other hi-fi entertainment tools in consumers' living rooms, multimedia interfaces have also been elevated to the top of their minds! One such high-speed multimedia interface that has become very popular now is

the high-definition multimedia interface (HDMI).

Using the earlier digital visual interface (DVI) as a starting point, HDMI provides high-speed transport of digital video and audio and operates at speeds in the range of 250 Mbps to 1.65 Gbps. The latest HDMI versions 1.3 and 1.4 extend data transfers rates to 3.4 Gbps.

HDMI connects digital audio/video sources such as set-top boxes, HD DVD players, Blu-ray disk players, PCs and game consoles to compatible digital audio devices, computer monitors and digital televisions. The interface supports on a single cable, any uncompressed television or PC video format, especially HD video, up to eight channels of compressed or uncompressed digital audio, and a consumer electronics control connection that enables devices to control each other. It is becoming the *de-facto* interface standard for HD televisions today.

Exciting future, with options unlimited

"Increase in digital information and new entertainment options are changing the consumers' expectations of a more satisfying and consistent real-time experience. This real-time experience has been enabled by the exponential growth in computing performance but is incomplete without providing equally fast access to content and data, no matter what the source. Faster delivery and response along with easier connectivity have created new technical requirements in capacity and speed. Servicing this need has established greater demands for increasing data exchanges at ever increasing bandwidths," comments Sun.

This rings true not only in the current generation of high-speed interfaces but as much in the research initiatives being taken up by various organisations towards a gigabit norm! ●

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